

TECHNOLOGY OVERVIEW AND DECISIONS

Principle: “Never Let the LLM fabricate a safety judgement, only real logic decides” Technologies Used:

1. Frontend: Flutter Dependencies: flutter_markdown, flutter_map

2. API Gateway: NodeJS Dependencies: express, dotenv, cors, axios, multer, firebase-admin

3. Authentication Layer: Firebase (Further Information by Team LEADER)

4. Backend: Dify

A. Nodes used: User Input Node, Agent Node, Code Node, Variable Aggregator Node, IF/ELSE Node, HTTP Request Node, Output Node, Answer Node

B. Python libraries used: json, gzip, datetime, typing, re, math

C. Flows Used: Chatflow: 1 (ORCA) Workflow: 4 (Weather, Ocean Analytics, Correlation, GeoSpatial Reasoning)

D. Model Used: Gemini (Flash-Lite)

E. Dify Plugins: Dify Agent Strategies[Its the official Strategies Collection by Dify: Currently 2 strategies are present Function Calling and ReAct we are using Function Calling]

5. Server for Sea route planning: Hosted on Render Python Libraries Used:

fastapi, uvicorn[standard], searoute, httpx, pydantic, shapely

6. API's used:

A. Bhashini API: functions (ASR, TTS, ALD{also called as LID})

B. (Dify Backend) Weather: Nominatim[location name to coordinates fetch and vice versa], Open Meteo[Weather Data fetch]

C. (Dify Backend) Ocean Analytics: Nominatim[location name to coordinates fetch and vice versa], Marine Open Meteo[Marine Data fetch], Erddap NOAA Coastwatch[Chlorophyll Data fetch], GDACS[Cyclone Data fetch]

7. Web Scrapping for Potential Fishing Zone: INCOIS offers PFZ data with static links:

1. Gujrat: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC001>
2. Maharashtra: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC002>
3. Goa: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC003>
4. Karnataka: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC004>
5. Lakshwadeep: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC014>
6. Kerla: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC005>
7. South Tamilnadu: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC006>
8. North Tamilnadu: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC007>
9. South AndraPradesh: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC008>
10. North AndraPrades: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC009>
11. Orrisa: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC010>
12. West Bengal: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC011>
13. Andaman: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC012>
14. Nicobar: <https://incois.gov.in/MarineFisheries/TextData?secid=SEC013>

End points:

IMPORTANT

1. We are currently using only 1 AI Model and 1 Bhashini API the rest are free API's that are free of cost
2. PFZ is scrapped through manual python code hence it also free of cost
3. FireBase is a security necessity which can be changed in future to a personalized authentication server hence saving us the cost of a third party provider and allowing us scalability

Design Decisions

The ORCA architecture was designed to satisfy the functional requirements of **SIH Problem Statement ID 26176**, including autonomous planning, collaborative reasoning, multi-source data integration, multilingual interaction, geospatial intelligence, and explainable decision support.

Collaborative Multi-Agent Architecture

Marine intelligence requires simultaneous analysis of weather, **Satellite Earth Observation**, **PFZ**, GIS, and hazard advisories. ORCA therefore adopts a **Collaborative Multi-Agent Architecture**, where specialized agents independently analyze domain-specific information while the **Planner Agent** coordinates task execution and synthesizes the final response.

LangGraph:

LangGraph was selected for agent orchestration because it supports **autonomous planning**, **task decomposition**, conversational memory, and structured multi-agent workflows.

Dify:

Dify manages conversational workflows, prompt orchestration, API integration, and AI application deployment, simplifying interaction between users and collaborative agents.

Flutter:

Flutter enables cross-platform mobile application development through a single codebase, ensuring accessibility for fishermen, researchers, and maritime authorities.

PostgreSQL + PostGIS:

Marine intelligence depends heavily on geospatial analysis. **PostgreSQL with PostGIS** provides efficient spatial indexing, geofencing, route optimization, and location-based querying required for marine decision support.

Gemini:

Gemini serves as the primary **Large Language Model (LLM)** responsible for natural language understanding, contextual reasoning, intent interpretation, and explainable recommendation generation.

Bhashini:

To support **Indian regional languages**, ORCA integrates **Bhashini**, enabling multilingual speech and text interaction and improving accessibility for coastal communities.

Table of Technology Selection Rationale

Requirement	Design Decision	Technology Selected	Rationale
Autonomous planning & task decomposition	Multi-Agent Architecture	LangGraph	Coordinates specialized AI agents and manages workflow execution
Conversational interaction	Workflow Management	Dify	Simplifies conversational workflow design and API integration
Natural language reasoning	Large Language Model	Gemini	Supports intent understanding and evidence-based response generation
Cross-platform accessibility	Mobile Application	Flutter	Single codebase with consistent cross-platform performance
Spatial reasoning & geofencing	Geospatial Database	PostgreSQL + PostGIS	Efficient storage and processing of geospatial datasets
Multilingual interaction	Language Processing	Bhashini	Enables regional language communication and accessibility